



PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF GEOGRAPHY

GEOGRAPHY

—PRELIM PAPER 2—

12 September 2024

Examiner: Mr. D. Grace

Internal Moderator: Mr. A Meintjes

External Moderator: Mr. A. Wegener

TIME: 3 hours

MARKS: 150 Marks

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. Write your name and class in the space provided on the answer book.
 2. Please check that your question paper is complete.
 3. Read the questions carefully and answer ALL the questions in your answer booklet.
 4. It is in your own interest to write legibly and to present your work neatly.
 5. You are provided with a 1 : 50 000 topographic map 3325DC & DD and 3425BA GQEBERHA and an orthophoto 3325 DC 23 GQEBERHA of a part of the mapped area.
 6. You must hand the topographic map and orthophoto map to the invigilator at the end of this examination session. You may only use pencil on these maps. Please erase any pencil markings on the map before handing it back in.
 7. Show ALL calculations and formulae, where applicable. Marks will be allocated for these.
 8. Indicate the unit of measurement or compass direction in the final answer of calculations, e.g. 10 km; 2.1cm; west of true north.
 9. You may use a non-programmable calculator.
 10. You may make use of a magnifying glass.
 11. The area demarcated in RED/BLACK on the topographic map represents the area covered by the orthophoto map.
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SECTION A – Settlement Geography

QUESTION 1: Short Questions

1.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

1.1.1. Urban settlements that are accessible and provide goods and services to surrounding areas are known as ...

- A. Specialised cities.
- B. Break-of-bulk points.
- C. Central places.
- D. Gateway towns. (1)

1.1.2. Settlements that are established at points where one mode of transport is exchanged for another are known as ...

- A. Specialised cities.
- B. Break-of-bulk points.
- C. Central places.
- D. Gateway towns. (1)

1.1.3. Vatican City is a settlement devoted almost entirely to religion. This is an example of what kind of settlement?

- A. Junction town.
- B. Central place.
- C. Gap town.
- D. Specialised city. (1)

1.1.4. Saldanha Bay is a settlement that was established around the largest and deepest natural port in South Africa. What type of settlement would this be?

- A. Break-of-bulk point.
- B. Junction town.
- C. Gateway town.
- D. Central place. (1)

1.1.5. A settlement that is established around mining operations would be primarily associated with which functions?

- A. Primary functions.
- B. Secondary functions.
- C. Tertiary functions.
- D. Quaternary functions. (1)

1.1.6. A settlement that is established around universities such as Oxford would be primarily associated with which functions?

- A. Primary functions.
- B. Secondary functions.
- C. Tertiary functions.
- D. Quaternary functions. (1)

1.1.7. Which of the following main ideas underpin Central Place Theory?

- A. Distance results in travelling costs, and consumers always wish to reduce travelling costs.
- B. Businesses aim to maximise profits by attracting the most customers.
- C. People want the cheapest price and convenience when shopping.
- D. All of the above. (1)

1.1.8. The area from which a business draws its customers is known as ...

- A. Range of customers.
- B. Range of goods.
- C. Sphere of customers.
- D. Sphere of influence. (1)

(8 x 1) (8)

1.2. Read the following statements and choose the appropriate word(s) in brackets to make the sentence TRUE. Write down only the question number (1.2.1 to 1.2.7) and the correct answer in the ANSWER BOOK, e.g. 1.2.8 town.

1.2.1. The smallest rural settlements are (hamlets/isolated farmsteads).

1.2.2. A (town/hamlet) is the smallest urban settlement.

1.2.3. When two or more large urban cities merge, a (conurbation/metropolis) is formed.

1.2.4. A (metropolis/megalopolis) is the main city within a particular region.

1.2.5. The (multiple nuclei/third world) city model is based on the idea of multiple 'centres' within a city.

1.2.6. Overcrowding and congestion is a (centrifugal/centripetal) force that drives people and business out of a city centre.

1.2.7. Functional convenience would be a (centrifugal/centripetal) force that attracts businesses and people to a particular area in a city.

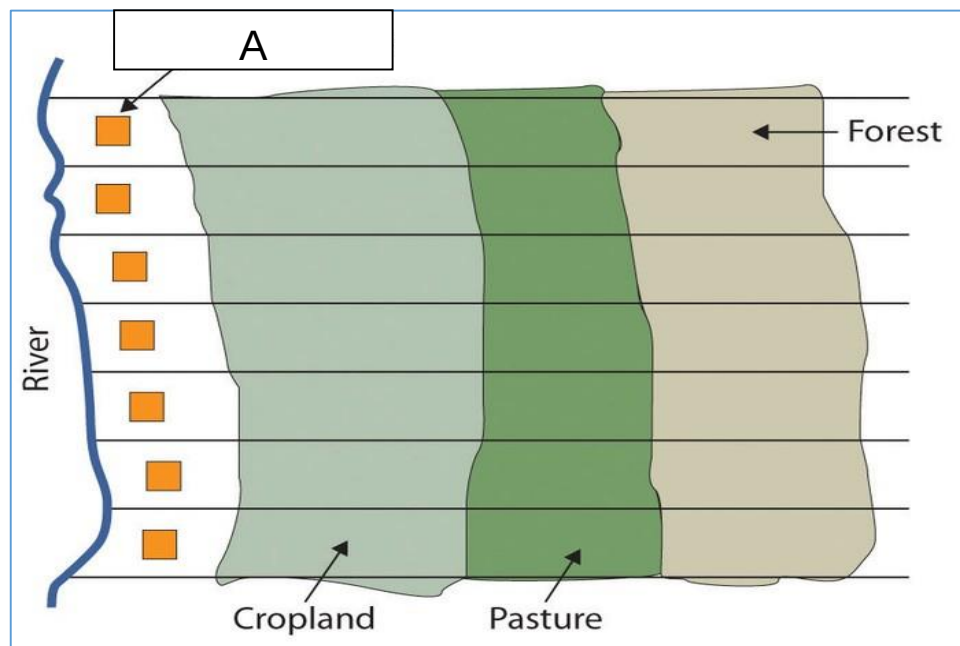
(7 x 1) (7)

[15]

1	C	1.1	Isolated farmstead
2	B	1.2	Town
3	D	1.3	Conurbation
4	A	1.4	Metropolis
5	A	1.5	Multiple Nuclei
6	D	1.6	Centrifugal
7	D	1.7	Centripetal
8	D		

QUESTION 2: Classification of Rural Settlements

2.1 Refer to the diagram below and answer the questions that follow:



2.1.1 **Is** the situation of the settlement at **A** positive or negative? (1x1)(1)
Positive

2.1.2 **Identify** the settlement pattern at **A**. (1x1)(1)
Linear

2.1.3 **Would** this settlement be a wet-point or dry-point settlement? (1x2)(2)
Provide a reason for your answer.
Wet point, it is near a river

2.1.4 **Describe TWO** factors that may have influenced the pattern and shape of the settlement labelled **A**. (2x2)(4)
The river: The settlement follows the river.

The relief: The gradient near the river is less steep and so would influence people to build there.

Roads: The building of roads to connect the settlement would also contribute towards pattern and shape.

- 2.1.5 **Differentiate** between the terms site and situation. (2x2)(4)

A site is the physical location of a settlement while the situation refers to the relationship between the settlement and its environment and surrounds.

- 2.1.6 **All** settlements have a function. The function relates to its economic and social development, and determines its main activities. **List three** potential functions of this settlement. (3x1)(3)

Fishing, Farming, Forestry, livestock.

[15]

QUESTION 3: Case study

- 3.1 Refer to the article below and answer the questions that follow:

Rural-urban migration in South Africa.

Besides the job opportunities that are a reality in urban areas, the migration to the towns and cities is a result of the neglect of rural communities in terms of service delivery. Urban areas are better supplied with social infrastructure than rural areas. As a result, people who can afford to buy or rent accommodation migrate with their families to urban areas. According to the United Nations projections, by 2030, South Africa's population living in urban areas will be 71.3%, with possibilities of increasing to 80% by 2050. While this could be good for economic development, it also goes with developmental imbalances between the urban and rural areas. The increase in rural-urban migration may prejudice the rural development through loss of skilled labour, prolonging the poverty and underdevelopment cycle.

According to Bello-Schunemann and Aucoin (2016), urban migration is regarded as a rational decision weighing projected advantages against costs. Rural-urban migration is therefore a response to factors that influence the desire for urban life over rural living, such as improved income, education quality, health services available in the city or negative income shocks in the rural areas. Urbanisation is a global phenomenon, with some 56% of the world's population residing in cities and the figure is expected to reach 68% by 2050 (OECD 2020). Urban migration is mainly driven by the economic prosperity in urban centres as people relocate to cities in search for employment or to take advantage of the urban market for trading and various economic activities. The World Bank (2021) indicates that over 80% of the gross domestic product (GDP) of the world is generated in cities, as factories and industries are mostly found in urban areas, hence a pulling factor for urbanisation (JOLGRI, 2024).

- 3.1.1 **What** do you understand by the sentence "Urban areas are better supplied with social infrastructure than rural areas"? (1x2)(2)

The sentence "Urban areas are better supplied with social infrastructure than rural areas" means that cities and towns generally have more and better-developed facilities and services that support the well-being of their populations compared to rural areas. Social infrastructure includes essential services such as healthcare (hospitals, clinics), education (schools, universities), public transportation, and recreational facilities. Urban areas tend to have more hospitals, schools, and public amenities due to higher population densities and greater economic resources, while rural areas often have limited access to these services, which can lead to disparities in quality of life between urban and rural regions.

- 3.1.2 **Distinguish** between rural depopulation and selective outward migration. (2x2)(4)

1. Rural Depopulation:

- **Definition:** This refers to the overall decline in the population of rural areas due to various factors, such as migration to urban areas, a lack of job opportunities, poor access to services, and changing agricultural practices.
- **Characteristics:** It often involves a broad segment of the population leaving rural areas, leading to deserted or shrinking villages and a reduction in the labor force, especially in agriculture. Rural depopulation can result in economic stagnation and a loss of essential services in those regions.
- **Example:** A small farming community experiencing a decrease in population as families move to cities for better job opportunities and living conditions.

2. Selective Outward Migration:

- **Definition:** This is a more specific type of migration where only certain groups of people, usually the young, skilled, or educated, leave a region (often rural) in search of better opportunities elsewhere, usually in urban areas or abroad.
- **Characteristics:** It does not necessarily result in a total population decline, but rather a demographic shift where the remaining population is often older and less economically active. Selective outward migration can lead to a "brain drain" in rural areas, leaving behind a

population that may struggle to maintain local economies.

- **Example:** Young adults leaving a rural area to pursue higher education or professional jobs in the city, while older residents remain in the countryside.

3.1.3 With reference to the article, **discuss** the **ONE** positive and **ONE** negative consequence of rural-urban migration. (2x2)(4)

Positive:

- More opportunities
- Access to more services
- Increased quality of life
- Greater earning potential.

Negative:

- Rural depopulation.
- Loss of skills

3.1.4 **Provide** and **explain ONE** cause of rural depopulation. (1x2)(2)

Crime, lack of opportunities, Selective outward migration. Any reasonable answer.

3.1.5 **List THREE** consequences of rural depopulation for the rural settlement. (3x1)(3)

Economic Decline

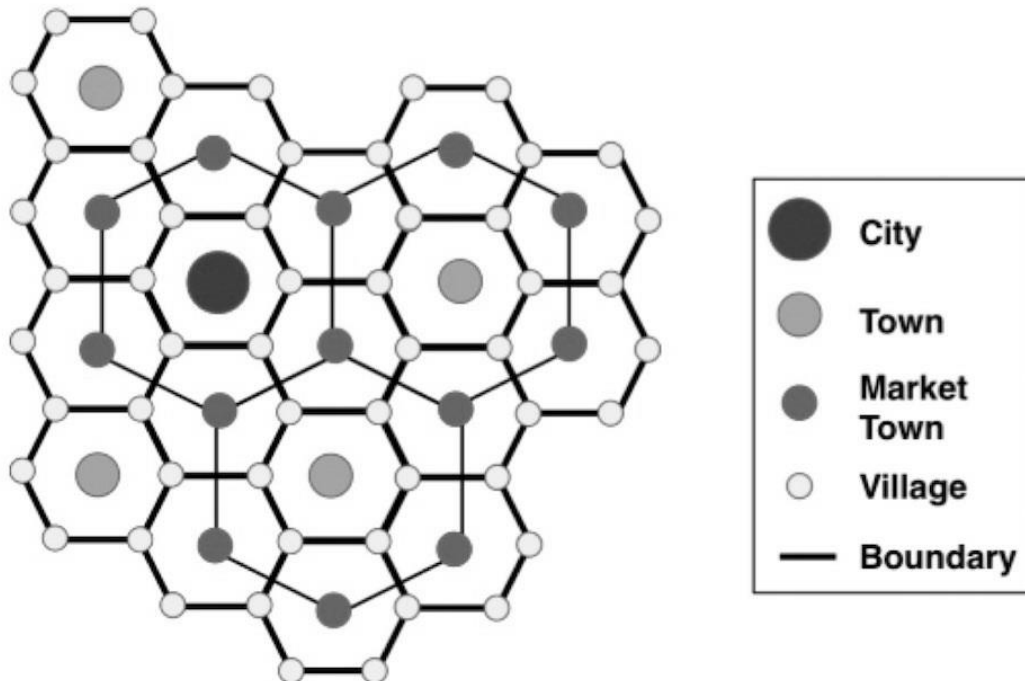
[15]

Aging Population

Loss of Services and Infrastructure

QUESTION 4: Central place theory

4.1 Refer to the diagram below and answer the questions that follow:



4.1.1 **What** is a conurbation? (1x2)(2)

A **conurbation** is a large urban area that forms when multiple cities, towns, or urban settlements expand and merge together, creating a continuous metropolitan region. Unlike individual cities, a conurbation consists of multiple municipalities or urban centres that have grown into each other due to population growth, urban sprawl, and infrastructure development, while still maintaining distinct identities.

4.1.2 Urban hierarchies are diagrammatical representations of the number of settlements in an area. **Draw** a pyramidal urban hierarchy for the diagram above. (1x2)(2)

City
Town
Market town
Village

4.1.3 According to Central Place theory, **do** we find more lower order places than higher order places in a country? (1x1)(1)

Lower order places.

4.1.4 **Provide** a reason for your answer to QUESTION 4.1.3. (1x2)(2)

There is more demand for lower order places as we use them more frequently.

- 4.1.5 In a paragraph of approximately eight lines. **Explain** how Walter Christaller's Central Place Theory helps in understanding the distribution of cities and towns in a region. In your answer, discuss the concepts of "threshold" and "range," and how they determine the hierarchy of settlements according to the theory. Use examples to illustrate how the theory applies to real-world urban planning and settlement patterns in South Africa. (4x2)(8) **[15]**

Walter Christaller's **Central Place Theory** explains how cities and towns are distributed based on their functions and the services they provide. The theory introduces two key concepts: **threshold** (the minimum population required to support a service) and **range** (the maximum distance people are willing to travel to access that service). According to this theory, larger cities with a high population have a higher threshold and offer more specialized services, while smaller towns have a lower threshold and provide more basic goods. For example, in South Africa, cities like Johannesburg serve as higher-order settlements offering specialized services such as financial institutions and large hospitals, whereas smaller towns like Harrismith offer more basic services such as grocery stores and clinics. This hierarchy helps in urban planning by determining where different types of services and infrastructure should be located based on the population distribution and demand in different regions. Dave's law.

TOTAL SECTION A: [60]

SECTION B – Economic Geography

QUESTION 5: Short Questions

5.1 Choose the word/term from COLUMN B that matches the statement in COLUMN A. Write only **Y** or **Z** next to the question numbers (5.1.1 to 5.1.7) in the ANSWER BOOK, e.g. **5.1.8 Y**.

COLUMN A	COLUMN B
5.1.1 Unfavourable trade balance ...	Y increases employment Z increases unemployment
5.1.2 Total value of all finished goods and services produced by a country's citizens in a given financial year, irrespective of their location.	Y Gross national product Z Gross domestic product
5.1.3 Difference in value between a country's imports and exports.	Y balance of trade Z balance of payment
5.1.4 A country's financial statement showing its transactions with the rest of the world is known as ...	Y balance of payment Z balance of trade
5.1.5 A trade ... exists in a country when the value of the imports is greater than the value of exports.	Y surplus Z deficit
5.1.6 Goods that are exchanged within a country is known as trade.	Y regional Z domestic
5.1.7 A silo is classified as being in the ...	Y primary sector Z secondary sector

(7x1) (7)

5.2. Various options are provided as possible answers to the following questions. Choose the answer and write the letter **A-D** next to the question numbers. (5.2.1 to 5.2.8 in the ANSWER BOOK, e.g. **5.2.9 D**).

5.2.1 A coal fired plant is an example of a ... industry.

- A market oriented.
- B raw material orientated.
- C bridge.
- D footloose.

5.2.2 An industry that can be in any place without being affected by factors such as resources or transport.

- A Footloose industry.
- B Ubiquitous industry.
- C Break of bulk industry.
- D Market orientated industry.

5.2.3 Industries that provide services that are available 24 hours a day 7 days a Week from any geographic location.

- A Market orientated industry.
- B Heavy industry.
- C Light industry.
- D Ubiquitous industry.

5.2.4 Heavy industries are characterised by ...

- (i) proximity to bulk transport facilities.
- (ii) being ubiquitous.
- (iii) high levels of noise and air pollution.
- (iv) being market orientated.

- A (i) and (ii)
- B (i) and (iii)
- C (ii) and (iii)
- D (iii) and (iv)

5.2.5 TWO types of industries generally associated with lower levels of air pollution.

- (i) Light industry.
- (ii) Raw material orientated industry.
- (iii) Ubiquitous industry.
- (iv) Heavy industry

- A (i) and (ii)
- B (ii) and (iii)
- C (i) and (iii)
- D (ii) and (iv)

5.2.6 An industry associated with perishable goods and is located close to the consumer.

- A Raw material orientated industry.
- B Light industry.
- C Market orientated industry.
- D Heavy industry.

5.2.7 TWO types of industries associated with bulk transport.

- (i) Ubiquitous industry.
- (ii) Light industry.
- (iii) Heavy industry.
- (iv) Raw material orientated industry.

- A (i) and (ii)
- B (i) and (iv)
- C (ii) and (iii)
- D (iii) and (iv)

5.2.8 The industry which is located between the raw material and the market

- A Bridge industry
- B Raw-material oriented industry
- C Market orientated industry
- D Heavy industry

(8x1)(8)

5.1.1	Z	5.2.1	B
5.1.2	Y	5.2.2	A
5.1.3	Y	5.2.3	D
5.1.4	Y	5.2.4	B
5.1.5	Z	5.2.5	C
5.1.6	Z	5.2.6	C
5.1.7	Y	5.3.7	D
		5.2.8	A

[15]

QUESTION 6: Agriculture

6.1 Refer to the extract below based on beef farming in South Africa.

Beef farming in South Africa

The beef industry in South Africa faces numerous challenges, continued loadshedding, global geopolitical tensions and the ongoing cost-of-living crisis. South Africa produces 83% of its meat requirements. The local demand for beef generally outstrips production, even though there are untapped reserves in the communal farming areas. In South Africa, beef production is characterised by its dual nature of small-scale and large-scale farming. Since cattle serves as an indicator of wealth in some communities, communal farming does not contribute to South Africa's meat requirements. Characteristically, beef production is a long-term undertaking, and profits are rarely made in the short term. The climatic conditions in some parts of South Africa also are not suitable for beef production.

6.1.1 **Quote TWO** pieces of evidence from the above extract that hinder beef farming in South Africa. (2x1)(2)

“Communal farming does not contribute to South Africa’s meat requirements”

“Climatic conditions in some parts of South Africa also are not suitable for beef production”

6.1.2 **What** is the percentage shortfall in South Africa regarding its production of meat requirements? (1x1)(1)
17 percent.

6.1.3 **Why** is beef production important for South Africa? (1x2)(2)
It is a source of food and is a indicator of wealth in some communities.

6.1.4 **Describe TWO** natural factors that have a negative impact on beef production in South Africa. (2x2)(4)

Drought: South Africa is prone to periods of severe drought, particularly in arid and semi-arid regions. Drought reduces the availability of water and grazing land for cattle, which can lead to poor health, reduced growth rates, lower reproduction rates, and even death of livestock. This limits beef production and

increases costs for farmers who may have to rely on expensive feed and water resources.

Diseases and Pests: Livestock in South Africa are susceptible to diseases such as foot-and-mouth disease, tick-borne illnesses, and bovine tuberculosis, which can lead to severe losses in beef production. These diseases can cause cattle to suffer from poor health or die, leading to a decrease in production. Pests like ticks, which thrive in certain climates, can also transmit diseases and further affect the health and productivity of cattle herds.

- 6.1.5 **Discuss** sustainable strategies that can be implemented by the South African government to assist small scale beef farmers to increase beef production.

(3x2)(6)

[15]

To assist small-scale beef farmers in South Africa and promote sustainable beef production, the government can implement the following strategies:

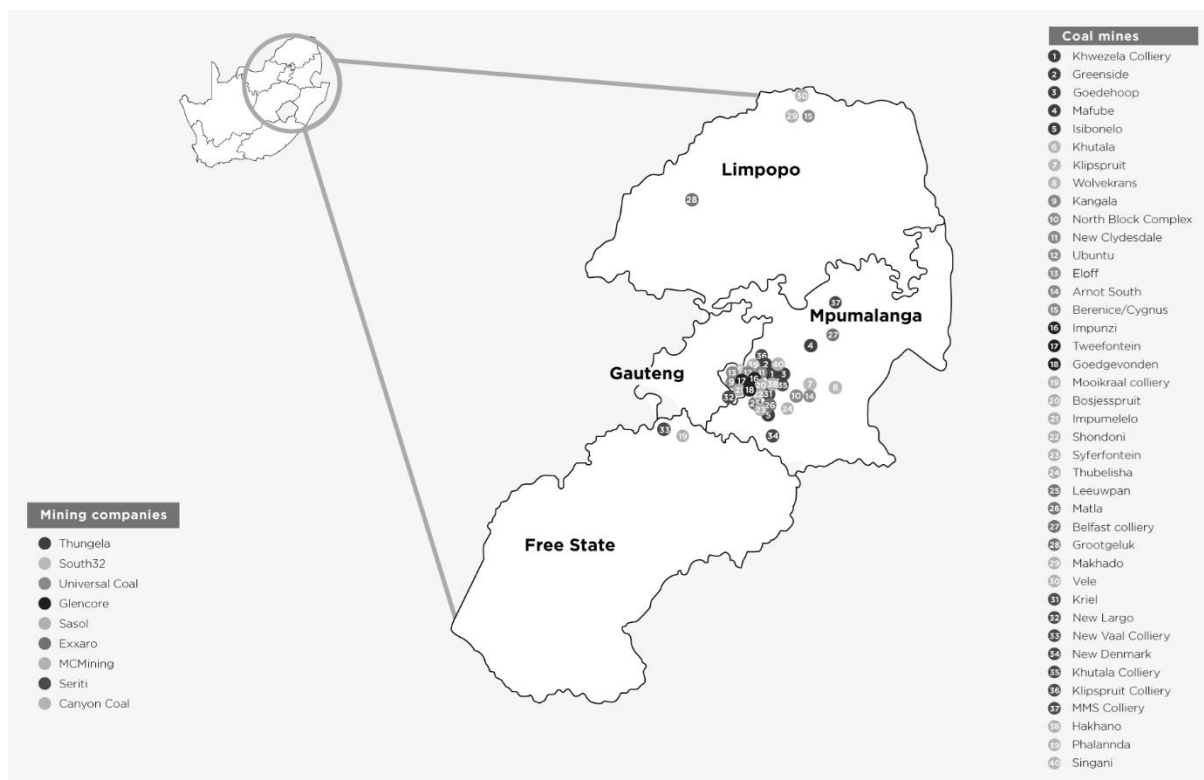
1. **Water and Grazing Management:** The government can provide training and resources on sustainable water and grazing practices. This includes educating farmers on rotational grazing, which allows the land to recover and prevents overgrazing, as well as the use of rainwater harvesting techniques to ensure a consistent water supply during dry periods. Additionally, introducing drought-resistant grasses can help maintain cattle feed during harsh weather conditions.
2. **Access to Veterinary Services and Disease Control:** The government should invest in expanding veterinary services and providing affordable access to vaccinations and treatments for livestock diseases. Implementing widespread tick control programs and offering disease-monitoring systems would reduce livestock losses due to preventable illnesses, ensuring healthier herds and increased beef production.
3. **Financial Support and Access to Markets:** Small-scale farmers often lack the financial resources to invest in better farming practices or access larger markets. The government can offer subsidized loans or grants to help farmers purchase equipment, improve infrastructure, and invest in quality feed. Additionally, establishing cooperatives or market hubs where small-scale farmers can collectively sell their beef would

enhance their bargaining power and provide easier access to local and national markets.

By combining these strategies, the South African government can help small-scale farmers boost production while maintaining environmental sustainability and economic viability.

QUESTION 7: Mining

7.1 Refer to the figure below showing a graph and answer the questions that follow:



7.1.1 Which province accounts for the highest coal production in South Africa?

(1x1)(1)

Mpumalanga

7.1.2 Name TWO of South Africa's major industries that are dependent on coal.

(2x1)(2)

SASOL, ESKOM

- 7.1.3 **Name and explain TWO** factors that have contributed positively to South African mining. (2x2)(4)

Rich Mineral Resources: South Africa is home to an abundance of valuable minerals, including gold, platinum, diamonds, coal, and iron ore. The country's rich geological deposits have made it one of the leading global producers of several minerals, which has historically attracted significant investment and contributed to the growth of the mining industry. For example, South Africa has the world's largest reserves of platinum and significant gold deposits, which have been key drivers of the country's economic growth and development.

Advanced Mining Technology and Infrastructure: Over the years, South Africa has developed sophisticated mining techniques and infrastructure that have enabled the extraction of minerals from deep and complex underground deposits. Technological advancements in drilling, extraction, and processing have improved mining efficiency and productivity. Additionally, a well-established network of railways, ports, and roads has facilitated the transportation of minerals to both domestic and international markets, further contributing to the success of the mining sector.

Resources are near the surface: easier to mine.

Low thermal gradient: easier to mine and greater profits.

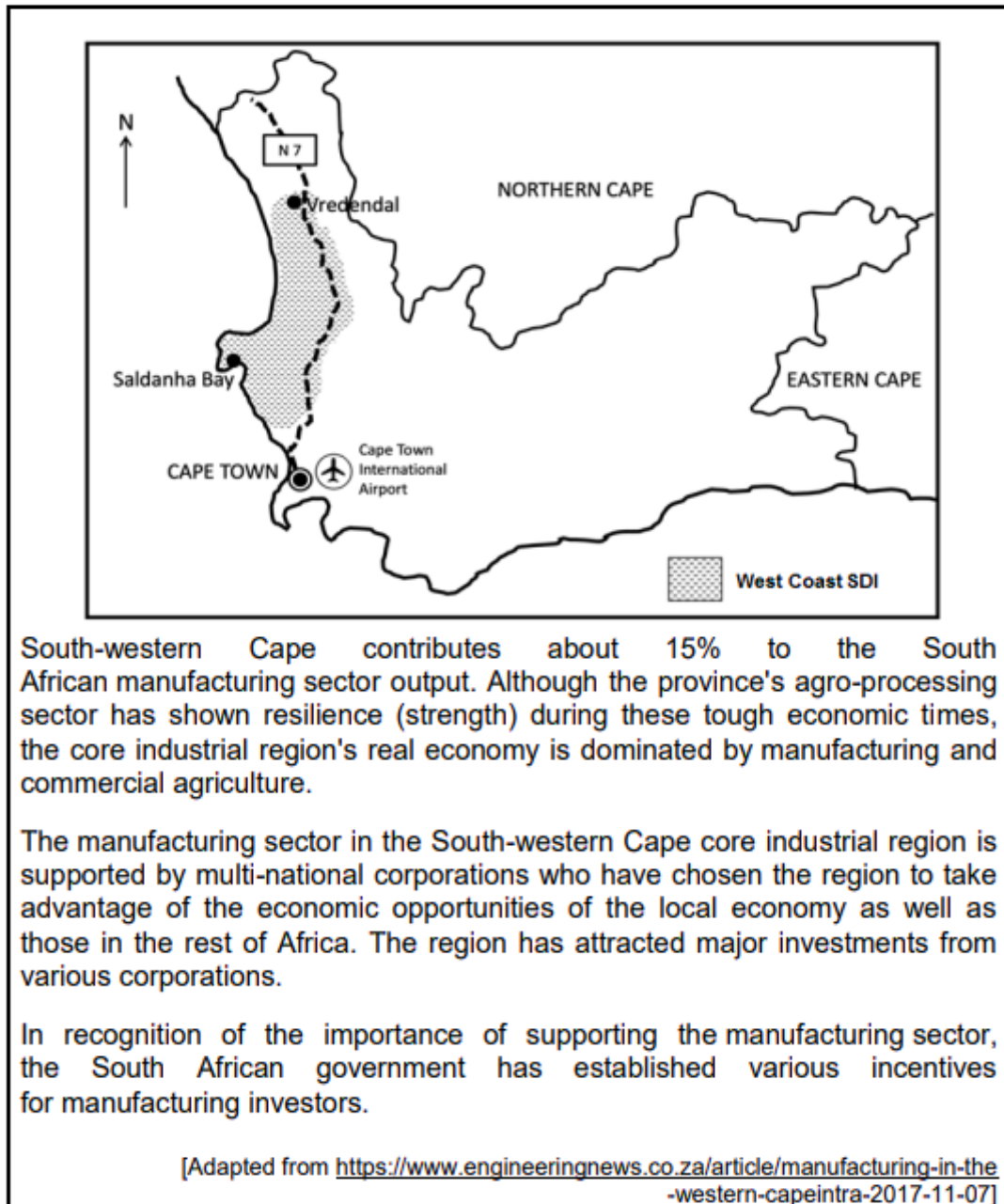
- 7.1.4 Mining and coal-dependant companies should work together to rectify environmental and social injustices associated with coal mining. In a paragraph of approximately eight lines, **explain** how this can be achieved. (4x2)(8)

Mining and coal-dependent companies can work together to rectify environmental and social injustices associated with coal mining by adopting a more sustainable and inclusive approach. They should invest in **rehabilitating damaged ecosystems** through land reclamation, reforestation, and reducing emissions caused by mining operations. These companies can also collaborate with local communities to provide **alternative livelihood programs** for displaced workers and residents affected by mining activities. Moreover, coal-dependent industries should transition to **cleaner energy sources**, reducing reliance on coal while ensuring affected workers are retrained and upskilled for employment in the renewable energy sector. By investing in **community** [15]

development projects such as education, healthcare, and infrastructure, they can address the long-term social impacts of mining. This collaborative approach promotes environmental sustainability and social equity.

QUESTION 8: Industrial Development Zones

8.1 Refer to the figure showing a map and answer the questions that follow:



INFORMATION: THE SALDANHA BAY INDUSTRIAL DEVELOPMENT ZONE (SBIDZ)

The largest and deepest (approximately 23 metres deep) natural port, at the most southern tip of Africa, located on the West Coast region of the Western Cape, is known as Saldanha Bay.

The development of South Africa's oil and gas resources as well as the fast growing of opportunities in Western Africa led to demand for an offshore oil and gas platform and components. Therefore, Saldanha Bay was the ideal choice due to the need for services and repair services to support these operations. The deep port of Saldanha Bay is politically and geographically the ideal location for this. Its location also creates a linkage point for both Sub-Saharan Africa and other international markets.

The SBIDZ's aim is to become a centralised provider of bulk parts and services to the industrial sector and home to many opportunities as an oil, gas and marine repair service hub.

8.1.1 **Define** the term *Industrial Development Zone*. (1x2)(2)

An **Industrial Development Zone (IDZ)** is a designated area, typically located near a port or major transportation hub, where specific policies and infrastructure are established to encourage industrial growth and investment.

8.1.2 **What** is the benefit of having a deep port? (1x2)(2)

Accommodating Large Vessels: Deep ports can handle larger ships, including large container ships, oil tankers, and bulk carriers, which require deeper water to dock and unload. This capability increases the port's capacity to manage higher volumes of cargo and facilitates international trade by allowing the direct docking of large vessels, reducing the need for transshipment.

8.1.3 **List ONE** factor that has limited the development of heavy industries in the South-western Cape core industrial region. (1x1)(1)

Infrastructure, distance. Any reasonable answer.

8.1.4 **Explain** how the West Coast Spatial Development Initiative (SDI) creates increased access for the South-western Cape core industrial regions to international markets. (1x2)(2)

The West Coast Spatial Development Initiative (SDI) enhances access for the South-Western Cape core industrial regions to international markets through several strategic measures. By focusing on developing key infrastructure and transportation networks, the SDI aims to improve connectivity and facilitate smoother logistics for businesses.

1. **Infrastructure Development:** The SDI invests in upgrading and expanding critical infrastructure, including roads, railways, and ports. For example, the Saldanha Bay deepwater port, a key component of the SDI, supports large-scale shipping operations and provides direct access to global trade routes. This deepwater capability allows larger vessels to dock, which is crucial for handling significant cargo volumes and reducing shipping costs.
2. **Transportation Efficiency:** Improved transportation networks, such as better roads and rail connections, enhance the movement of goods between industrial regions and the port. This increased efficiency reduces transit times and transportation costs, making it easier for businesses in the South-Western Cape to export products to international markets.
3. **Economic Zones and Incentives:** The SDI often includes the establishment of industrial development zones, such as the Saldanha Bay IDZ, which offer business incentives and streamlined regulations to attract investment. These zones are strategically located to leverage proximity to ports and transport links, further facilitating international trade.

- 8.1.5 In a paragraph of approximately eight lines, **explain** how the Saldanha Bay Industrial Development Zone has positively impacted the industrial growth and economic development of the South Western Cape. You may include specific examples of its contributions to local industries, job creation, and regional infrastructure.

(4x2)(8)

The Saldanha Bay Industrial Development Zone (IDZ) has significantly boosted industrial growth and economic development in the South Western Cape by providing a strategic location with advanced infrastructure and business incentives. The IDZ has attracted major investments in industries such as steel manufacturing, ship repair, and logistics, fostering local industrial growth. For instance, the zone hosts the Saldanha Steel Plant, which has bolstered the steel industry and supported regional manufacturing. Additionally, the IDZ has created numerous job opportunities for local residents, enhancing employment prospects and economic stability. Improved infrastructure, including a deepwater port and efficient transport links, has further facilitated trade and attracted additional businesses. Overall, the Saldanha Bay IDZ has played a crucial role in stimulating economic development and strengthening the industrial base of the South Western Cape.

[15]

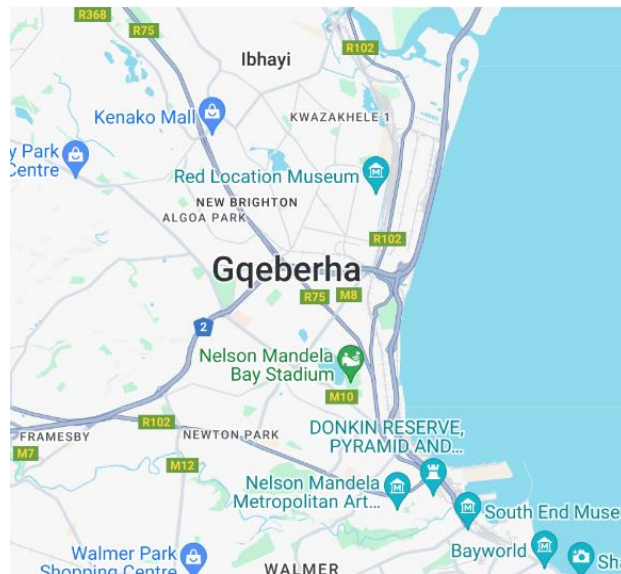
TOTAL SECTION B: [60]

SECTION C - Mapwork

RESOURCE MATERIAL

9.1 3325DC & DD and 3425BA GQEBERHA and an orthophoto 3325 DC 23
GQEBERHA

GENERAL INFORMATION ON GQEBERHA



Gqeberha (Xhosa: [\[g!ḥébéːxà\]](#); English: [/kəˈbærxə/](#) *keh-BAIR-khə*), formerly known as **Port Elizabeth**, and colloquially referred to as **P.E.**, is a major seaport and the most populous city in the [Eastern Cape](#) province of [South Africa](#). It is the seat of the [Nelson Mandela Bay Metropolitan Municipality](#) and home of the Nelson Mandela Bay Stadium. It is also South Africa's second-smallest metropolitan municipality by area. It is the sixth-most populous city in South Africa and is the cultural, economic and financial hub of the Eastern Cape.

Located on the western portion of [Algoa Bay](#) along the southeastern coast of South Africa, the city lies 770 km east of Cape Town. It is east of the [Garden Route](#) and faces the [Indian Ocean](#). It covers 251 square kilometers of the Nelson Mandela Bay metropolitan area, and is administered by South Africa's sixth-largest metropolitan municipality. The city's warm [oceanic climate](#) ranks it among the top cities in the

world for pleasant year-round weather. The city is known for many [blue-flag beaches](#) along the city's urban coastline; its popularity as an international and local holiday destination; and its rich and diverse cultural heritage. It is a gateway city for the Eastern Cape's adventure, outdoor and African [big five game](#) safari tourism.

QUESTION 9: Calculations

9.1.1 Provide the grid reference of the map directly South of **3325DC**. (1x2)(2)

9.1.2 **3425BA**
Calculate the straight line distance of the landing strip in block E5 in km. (1x2)(2)

Use the formula:

$$\frac{\text{Distance on a map} \times \text{scale}}{100\ 000}$$

$$3.4\text{ cm} \times 0.5 = 1.7\text{ km}$$

9.1.3 **Draw** a simple cross section between trigonometrical beacon 380 in block A1 and trigonometrical beacon 10 in block B1. (1x2)(2)

9.1.4 **Calculate** the gradient from trigonometrical beacon 266 in block D3 to trigonometrical beacon 380 In block A1. (4x1)(4)

Use the formula:

$$\text{Gradient} = \frac{VI}{HE}$$

[10]

$$VI = 58.6\text{ m}$$

$$HE = 6.6\text{ km} = 6600\text{m}$$

$$G = = \frac{58.6}{6600}$$

$$1:112,62$$

QUESTION 10: Interpretation

- 10.1 In **which** province is Gqeberha located? (1x1)(1)
Eastern Cape
- 10.2 **What** type of economic activity is predominate in C5? **Provide** a reason for your answer. (2x1)(2)
Break of bulk/transport. There is railway lines and a harbour.
- 10.3 **What** is the dominant street pattern in block A1 of the Topographic map? **List ONE** benefit of this pattern. (2x1)(2)
Planned irregular. Better traffic flow, less traffic lights.
- 10.4 **Account** for the location of the firebreak in block D2. (1x2)(2)
To protect the houses and golf course from fire.
- 10.5 **Which** area is wealthier between Sidwell block A3 and Mill Park block C2 and C3? **Provide** a reason for your answer. (2x1)(1)
Mill park is wealthier. There is a golf course nearby.
- 10.6 Urbanisation has been increasing in Gqeberha lately. **Provide ONE** piece of evidence that supports this statement and **ONE** suggestion to further increase urbanisation in the future. (2x2)(4)

[12]

Evidence: One piece of evidence supporting the increase in urbanization in Gqeberha is the notable growth in residential and commercial development projects. Recent reports indicate a rise in new housing estates, shopping centres, and business hubs, reflecting the expanding urban infrastructure and increased demand for urban living spaces.

Suggestion: To further increase urbanization in Gqeberha, the city could invest in enhancing its public transportation network. Developing a comprehensive and efficient public transit system, including bus rapid transit (BRT) and improved rail connections, would make commuting more accessible and attractive for residents and businesses. Improved transportation infrastructure can support further urban growth by connecting residential areas with business districts, thereby stimulating economic development and encouraging more people to move to the city.

QUESTION 11: GIS

Consider the Orthophoto map.

11.1 **Is** feature 7 on the orthophoto map and example of a point, line or polygon feature? (1x1)(1)

Polygon

11.2 **Which** format, raster or vector, would be best for showing categories of data such as elevation? (1x1)(1)

Raster

11.3 **Name** the GIS concept that assigns a value or characteristic to each geographical feature. (1x1)(1)

Attribute data.

11.4 **Name ONE** layer that would have likely been considered by GIS consultants before developing the airport in blocks D2. (1x1)(1)

Elevation, Vegetation, Land usage.

11.5 **Explain** how you could use GIS in managing the buffer zone around the airport on this map. (2x2)(4)

Mapping and Analysis: GIS can create detailed maps of the buffer zone around the airport, highlighting key areas such as noise-sensitive zones, restricted land use areas, and environmental protection zones. By layering data such as land use, population density, and noise contours, planners can analyze the impact of airport operations on surrounding areas and make informed decisions about land use and development. **[8]**

Monitoring and Regulation: GIS allows for real-time monitoring of the buffer zone to ensure that land use complies with airport regulations and safety standards. By integrating satellite imagery and remote sensing data, GIS can help track changes in land use, identify unauthorized developments, and assess their impact on airport operations and nearby communities.

Planning and Zoning: GIS can assist in the planning and zoning process by identifying suitable areas for different types of land use within the buffer zone. For example, it can help determine where to place noise barriers, where residential development should be limited, or where commercial activities can be promoted. GIS tools can model various scenarios and predict the potential impact of different land use strategies.

Public Communication: GIS can be used to create visualizations and maps that communicate buffer zone regulations and planning decisions to the public. Interactive maps and dashboards can provide stakeholders with clear information about land use policies, noise levels, and other relevant factors, facilitating better understanding and engagement.

TOTAL SECTION C: [30]

GRAND TOTAL:
150 MARKS